Design and Evaluation of an Agentic AI-Driven Decision Support System for Explainable Early Disease Risk Detection in Companion Animals

Presented by: MTE Ranasinghe
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Supervisor - Dr. (Mrs) N Wedasinghe
Co-Supervisor - Mr. BGL Balasuriya

OVERVIEW

INTRODUCTION

- Companion animals often exhibit early disease symptoms that are subtle, non-specific, and easily overlooked by pet owners.
- Delayed recognition of health risks leads to late veterinary intervention, increased treatment costs, and reduced animal welfare.
- There is a need for an accessible, text-based AI system that supports early disease risk detection and preventive care for companion animals.

AIM

- To design, develop, and evaluate an Agentic AI-driven explainable decision support system for early disease risk detection in companion animals.

OBJECTIVES

1. To identify early disease risks using owner-observed symptoms and behavioural pattern.
2. To develop machine learning models for disease category prediction and risk stratification.
3. To implement an Agentic AI layer for autonomous reasoning, urgency assessment, and action recommendation.
4. To provide explainable AI outputs including confidence scores and symptom influence factors.
5. To integrate nearby clinic discovery, veterinary availability, and appointment booking.
6. To support preventive care through vaccination tracking and intelligent reminders.

RESEARCH GAP

- Existing pet health applications mainly provide reminders, static information, or reactive support. They lack autonomous reasoning, early risk prediction, and context-aware preventive guidance, especially in text-based and locally relevant systems.

EXPECTED OUTCOME

- A web mobile application that provides early disease risk alerts, preventive recommendations, vaccination reminders, nearest location with surgeons and intelligent veterinary guidance through Agentic AI.

LITERATURE ANALYSIS

Topic	Pros	Cons	Key Findings	Identified Research Gap
AI in Animal Disease Prediction (Shinde et al., 2025)	Provides overview of AI methods	Focuses on surveillance, not owner-level tools	AI improves early detection using data and sensors	Lack of mobile, agent-based decision support for pet owners
ML in Animal Healthcare (Das et al., 2024)	Reviews ML techniques for disease detection	Mostly theoretical	ML improves prediction accuracy	No real-world mobile implementation for pet owners
Veterinary AI Decision Support (Bouchemla et al., 2023)	Shows feasibility of AI in vet medicine	Clinical-focused	AI supports diagnosis and decision-making	No preventive, owner-accessible risk systems
AI Chatbots in Pet Health (Jokar et al., 2024)	Improves owner interaction	Limited autonomous reasoning	Chatbots can guide owners	No agentic decision-making or action automation
Sensor-based Animal Monitoring (Aguilar-Lazcano et al., 2023)	Uses behavior data	Hardware dependent	Behavior signals indicate health	No mobile symptom-based approach
ML Disease Risk Prediction (Kulkarni et al., 2025)	Real-time outbreak forecasting	Livestock-focused	ML predicts disease risk	Not designed for companion animals or owners
AI Veterinary Assistance (Jin et al., 2024)	Automates clinical symptom extraction	Clinical records required	AI improves vet workflow	No owner-level symptom collection
Dog Health Score System (Kim & Kim, 2024)	Quantitative health assessment	Limited reasoning	Health scores correlate with vet diagnosis	No agentic reasoning or action guidance
PAWPAL System (Aishwarya et al., 2025)	Combines image + NLP symptom analysis	Image-focused	AI supports early guidance	Lacks agentic AI, vaccination tracking, and real-time vet discovery
AI in Veterinary Imaging (Pereira et al., 2023)	Improves diagnostic imaging	Requires imaging data	AI enhances diagnosis accuracy	Not suitable for early behavioral symptom analysis
Predictive Epidemiology in Animals (Akinsulie et al., 2024)	Supports early intervention	Research-level	AI predicts disease trends	No mobile preventive decision support tools
ML-Based Health Monitoring Systems (Rathi & Sumathi, 2025)	Improves disease classification	Limited integration	ML supports health prediction	No autonomous reasoning or preventive care focus

FUNCTIONAL REQUIREMENTS

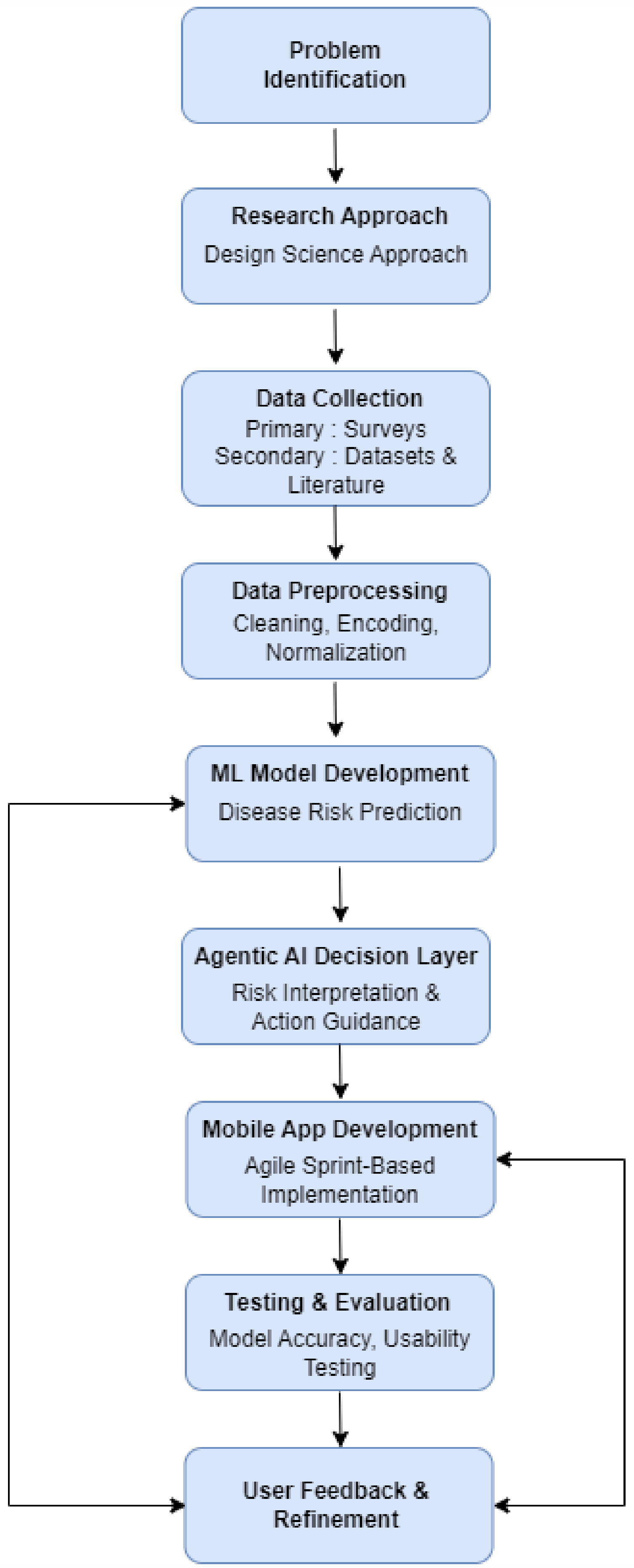
- User registration and secure login
- Pet profile creation and management
- Early disease risk prediction using machine learning
- Agentic AI-based decision support and guidance
- Veterinary clinic and surgeon discovery with time slots
- Vaccination tracking and reminder alerts
- Emergency risk alerts and referrals
- Health history management
- User notifications and feedback system

NON-FUNCTIONAL REQUIREMENTS

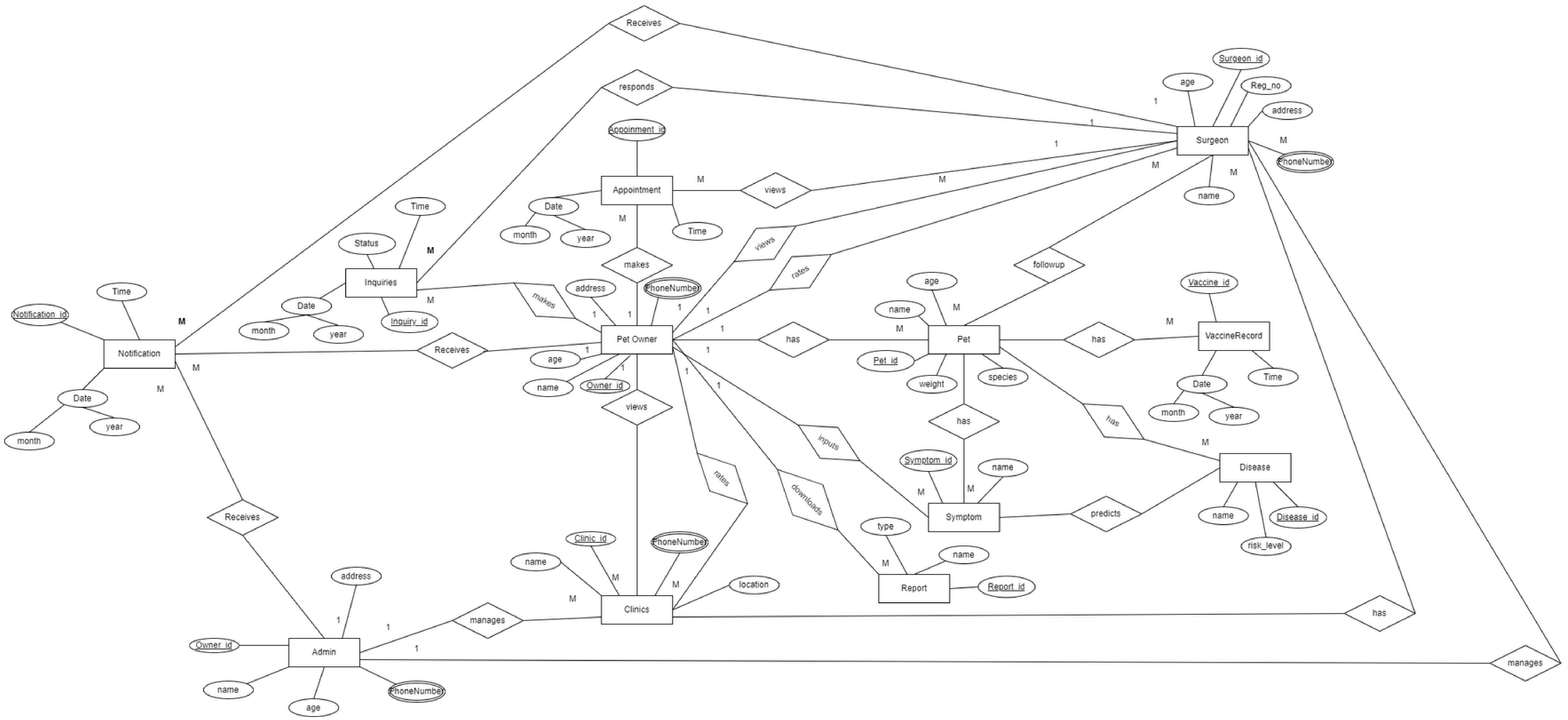
- Performance
- Scalability
- Usability
- Reliability
- Security
- Maintainability

METHODOLOGY

- Design Science Research approach with Agile development.
- Primary and secondary data used for ML training.
- Agentic AI layer with MCT model and ACL enables autonomous decision-making and action guidance.



ER DIAGRAM



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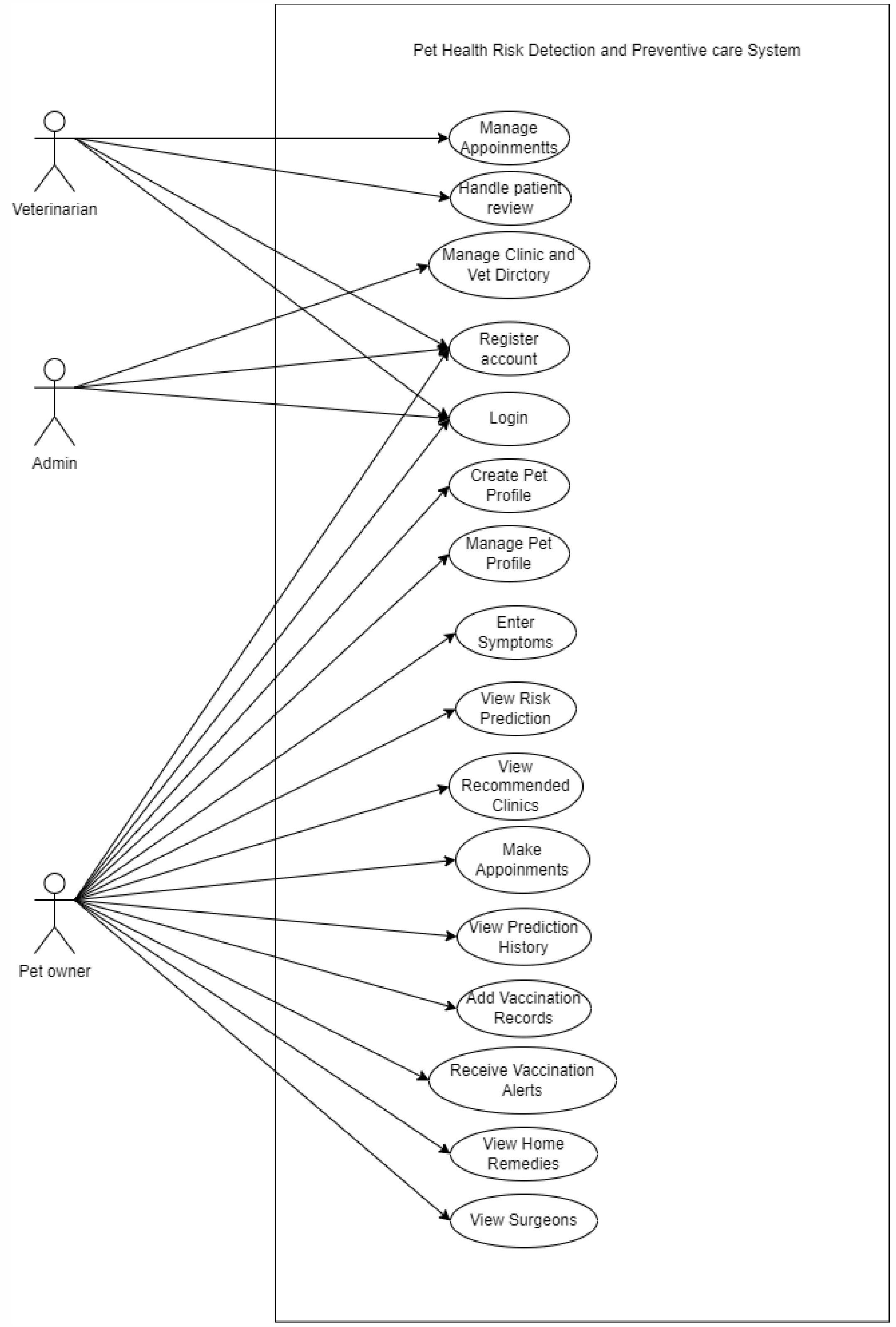
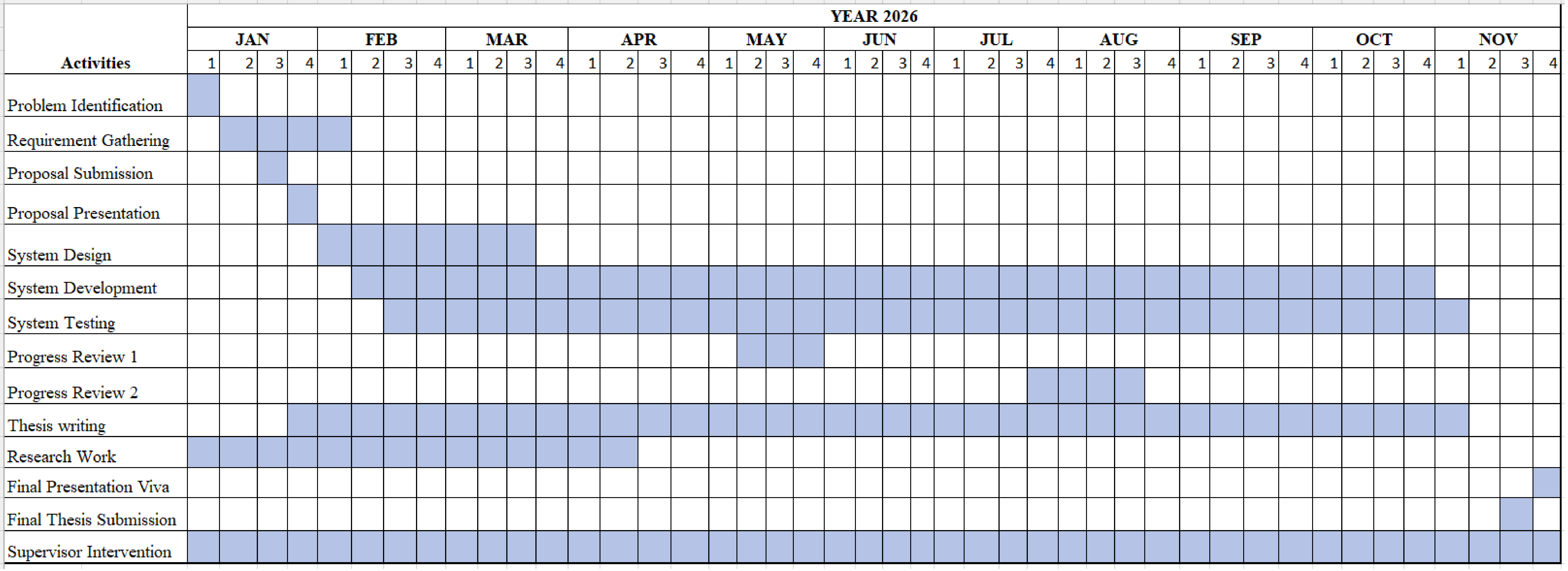
USECASE DIAGRAM



TECHNOLOGY

- Frontend: React.js (Mobile-responsive web UI)
- Backend: Node.js with Express.js
- Database: MongoDB
- Artificial Intelligence: Python Fast API, Google Vertex AI, Langchain, LangGraph
- Agentic AI Layer: LangGraph, Langsmith Graph
- Maps & Location: Google Maps API
- Notifications: Firebase Cloud Messaging

TIME PLAN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Activities
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Problem Identification
Requirement Gathering
Proposal Submission
Proposal Presentation
System Design
System Development
System Testing
Progress Review 1
Progress Review 2
Thesis writing
Research Work
Final Presentation Viva
Final Thesis Submission
Supervisor Intervention

Work Completed So Far	**Work to be Completed**
Research problem identification	Backend Implemetation / API integration
Proposal submission and approval	ML model training and evaluation
Literature review completion	Agentic AI workflow integration
Requirement gathering and product backlog	Google Maps clinic discovery
UML diagrams completed	Appointment module
Initial wireframes and UI flows	Admin analytics dashboard
Initial Implementation (Login and Registration, Pet Profile Management)	User testing and PR2 validation
Frontend implementation	Final dissertation and viva preparation

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Thank you for your time and consideration